2024

AUSTRALIAN MATHEMATICS COMPETITION

AMC.

AUSTRALIAN MATHS TRUST

Level C
Junior
Years 7–8

DATE
28 Sept.

TIME ALLOWED
75 minutes

Information 考试信息

- There are 30 questions: 25 multiple-choice and 5 integer.
  本次考试共 30 题，包含 25 道选择题和 5 道答案为整数的填空题。
- You are allowed 75 minutes to do the questions. There is no extra reading time.
  考试时长为 75 分钟，无额外读题时间。
- There are no penalties for incorrect answers. You should attempt all questions.
  答错不扣分，鼓励学生尝试解答所有问题。
- The questions have been checked thoroughly. Each question is clearly written and no further explanation will be given.
  本场考试的所有问题已仔细核查。题目陈述清晰，不提供更多解释。
- Diagrams are not drawn to scale. They are intended only as aids.
  图形未按比例绘制，仅供参考。

Competition rules 考试规则

- You must maintain silence at all times.
  考试过程中请保持安静。
- Do not open the paper until told to do so.
  收到监考老师指示前，请勿打开试卷。
- Mobile phones and smart watches are not allowed. If you have one you must give it to a supervisor, who will store it in a safe place.
  禁止使用手机和智能手表。如有携带这些设备，请立刻交与监考老师保管。
- You may use printed language dictionaries.
  允许使用纸质语言词典。
- You may NOT use calculators.
  不允许使用计算器。
- You may not borrow equipment without a supervisor's permission.
  未经监考老师许可，禁止向他人借用工具。
- You must not leave your seat during the competition. You must remain seated until the competition finishes.
  考试期间禁止离开座位，直到考试结束。
- If you wish to leave the room, a supervisor must accompany you. Raise your hand and wait for a supervisor to come to you.
  如需离开考场，必须举手示意并等待监考老师陪同。
- If you have any other questions or problems, raise your hand and wait for a supervisor.
  如果有其他问题，请举手示意监考老师。

No student may sit the competition more than once or sit more than one division of the competition. AMT conducts integrity checks on competition results and reserves the right to withhold results or disqualify students if plagiarism or duplicate sittings are suspected.

IMPORTANT

All papers must be immediately returned to a supervisor at the end of the competition.
考试结束后，由监考老师回收所有试卷。

This paper is not for circulation or distribution until Monday 30 September 2024.
在 2024 年 9 月 30 日星期一之前禁止讨论或传播试卷内容。

Copyright © 2024 Australian Mathematics Trust | ACN 083 950 341

Junior Division

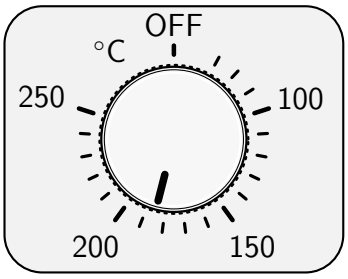
Questions 1 to 10, 3 marks each
1-10 题，每题 3 分

1. The metal in an Australian 20 cent coin is one-quarter nickel, and the rest is copper. What fraction is copper?
一枚澳大利亚 20 分硬币所含的金属成分中，有四分之一是镍，其余为铜。请问铜的占比是多少？

(A) $\frac{3}{4}$ (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $\frac{4}{4}$ (E) 0

---

2. Owen set the temperature dial on his oven as shown. What temperature is it set to?
小欧将烤箱温度调至如图所示的位置。请问他设定的温度是多少？



(A) 175 °C (B) 185 °C (C) 195 °C (D) 205 °C (E) 215 °C

---

3. If the day before yesterday was Monday, then the day after tomorrow will be
如果前天是星期一，那么后天将是

(A) Monday 星期一 (B) Tuesday 星期二 (C) Wednesday 星期三 (D) Thursday 星期四 (E) Friday 星期五

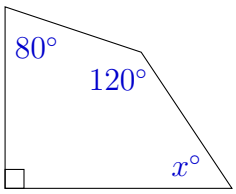
---

4. The value of $\frac{2024}{2+0+2+4}$ is
请问 $\frac{2024}{2+0+2+4}$ 的值是多少？

(A) 1 (B) 126.5 (C) 253 (D) 506 (E) 1012

---

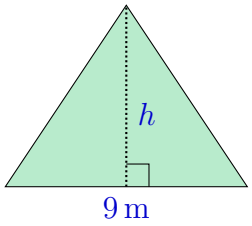
5. What is the value of x in the diagram?
请问图中 x 的值是多少？



(A) 40 (B) 50 (C) 60 (D) 70 (E) 80

---

6. The area of the triangle shown is $36\,\mathrm{m}^2$. What is the length h?
图中所示三角形的面积是 $36\,\mathrm{m}^2$。请问三角形的高 h 是多少？



(A) 2 m (B) 4 m (C) 4.5 m (D) 8 m (E) 16 m

---

7. What is the difference between $3 \times (4+2)$ and $(3 \times 4) + 2$?
请问 $3 \times (4+2)$ 与 $(3 \times 4) + 2$ 相差多少？

(A) 0 (B) 2 (C) 4 (D) 6 (E) 8

---

8. What number must be divided by 100 to get 0.2?
请问下列哪个数字除以 100 的结果为 0.2？

(A) 0.002 (B) 0.02 (C) 0.2 (D) 2 (E) 20

---

9. The average of 2024 and an unknown number is $2\frac{1}{2}$. What is the unknown number?
2024 与一个未知数的平均数为 $2\frac{1}{2}$。请问这个未知数是多少？

(A) 2019 (B) -2019 (C) 2029 (D) -2029 (E) -2024

---

10. The *Balga* is a grass tree found in south-western Australia. Balgas grow between 1 and 2 centimetres taller each year. Approximately how old is a Balga that is 4 metres tall?
巴尔加是一种生长在澳大利亚西南部的草树，每年可以长高 1 cm 到 2 cm。请问一棵巴尔加草树大约需要多少年才能长到 4 m？

(A) 10 years 10 年
(B) 30 years 30 年
(C) 60 years 60 年
(D) 300 years 300 年
(E) 1000 years 1000 年

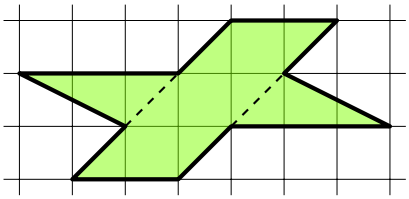
Questions 11 to 20, 4 marks each
11-20 题，每题 4 分

11. Danilo's pasta recipe requires one 60-gram egg for each 100 grams of flour, and no other ingredients. To make 800 grams of pasta, how many eggs does he need?
小洛的意面食谱上写明：每 100 g 面粉搭配一个 60 g 的鸡蛋，且无需其他配料。请问制作 800 g 的意面需要多少个鸡蛋？

(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

---

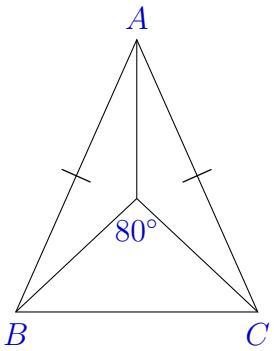
12. What is the area of the shape drawn in the grid?
请问网格中图形的面积是多少？



(A) 7 (B) 8 (C) 9 (D) 10 (E) 11

---

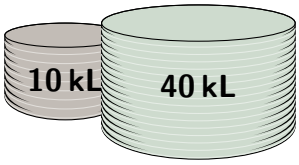
13. The isosceles triangle ABC is divided into three smaller isosceles triangles as shown. What is the value, in degrees, of $\angle ABC$?
如图所示，等腰三角形 ABC 被分割成三个较小的等腰三角形。请问 $\angle ABC$ 是多少度？



(A) 50 (B) 55 (C) 60 (D) 65 (E) 70

---

14. We have two water tanks for bushfire emergencies. The 10 kL tank is 80% full and the 40 kL tank is 30% full. Regulations state that we must have at least 30 kL available. How much more water do we need to add to our tanks?



我们有两个用于扑救森林火灾的水箱。容量为 10 kL 的水箱装有 80% 的水，而容量为 40 kL 的水箱装有 30% 的水。根据规定，我们必须保证至少有 30 kL 的水可供使用。请问还需要向水箱中加多少水才符合规定？

(A) 0 kL (B) 5 kL (C) 10 kL (D) 15 kL (E) 20 kL

---

15. I have a pad of 100 sticky notes. Each note is a 75 mm × 75 mm square. Using most of the pad, I completely cover all the faces of a cube. None of the notes folds over an edge of the cube. What is the largest side length of the cube could be?

我几乎用完一本数量为 100 张，尺寸为 75 mm × 75 mm 的正方形便签纸。使用过的便签纸可以完全覆盖一个立方体的所有表面，且没有便签纸在这个立方体的边缘处折叠。请问该立方体边长的最大可能值是多少？

(A) 22.5 cm (B) 30 cm (C) 45 cm (D) 60 cm (E) 75 cm

---

16. In this square, all rows and columns have the same sum. Find the average of A, B, C and D.

在这个正方形方阵中，每一行和每一列的元素之和相等。请计算 A、B、C、D 这四个数的平均值。

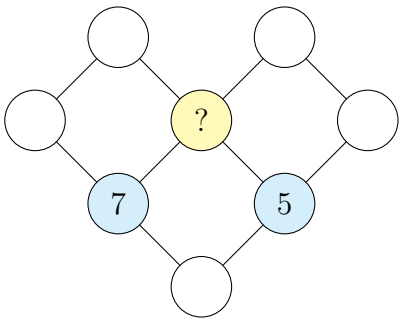
A	11	B
X	5	Y
D	7	C

(A) 7 (B) 8 (C) 9 (D) 10 (E) 11

---

17. In this puzzle, the numbers from 1 to 8 are to be placed in the 8 circles. For each of the 3 squares, the sum of the 4 numbers on the corners must be 20. With 5 and 7 placed as in the diagram, what number goes in the middle circle?

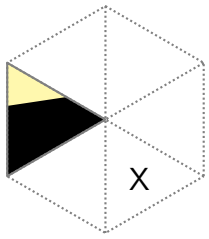
这道谜题中，你需要将数字 1 到 8 放置在 8 个圆圈里。图中有 3 个正方形，每个正方形 4 个角上的数字之和必须是 20。数字 5 和 7 的位置如图所示，请问中间的圆圈应该填入什么数字？



(A) 1 (B) 2 (C) 3 (D) 4 (E) 6

---

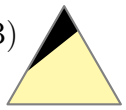
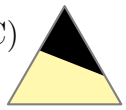
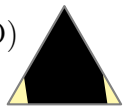
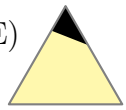
18. A regular hexagon made from yellow cardboard has a thick black stripe painted across it. The hexagon is cut up into six equilateral triangles and the pieces are mixed up. This diagram shows one of the pieces in its original position.



一条粗黑条纹横穿一个由黄色纸板制成的正六边形。将该正六边形切割成 6 个等边三角形碎片，然后打乱顺序。图中显示了其中一个三角形的原始位置。

The other five pieces are shown in options (A) to (E). Which piece was in the position marked with the X?

选项（A）到（E）中展示了其余 5 个三角形。请问下列哪个三角形原本处于标记为 X 的位置上？

(A)
(B) 
(C) 
(D) 
(E) 

---

19. A three-digit number has all digits nonzero. These digits are multiplied together and then this product is subtracted from the original number. What is the largest possible value of this difference?

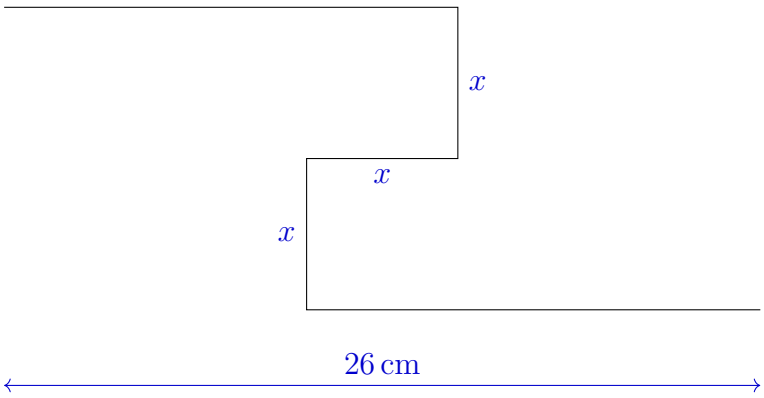
一个三位数，其各位数字均不为 0。将其各位数字相乘，然后用原数减去所得乘积。请问差值的最大可能值是多少？

(A) 890 (B) 901 (C) 909 (D) 910 (E) 990

---

20. A piece of wire of length 50 cm is bent into the shape shown, where each bend in the wire forms a right angle. The three internal sections of the bent wire are each the same length, x cm. What is the value of x?

一根 50 cm 长的铁丝被弯曲成如图所示的形状，其中每个弯曲处都构成一个直角。弯曲后的铁丝中有三段长度均为 x cm 。请问 x 的值是多少？

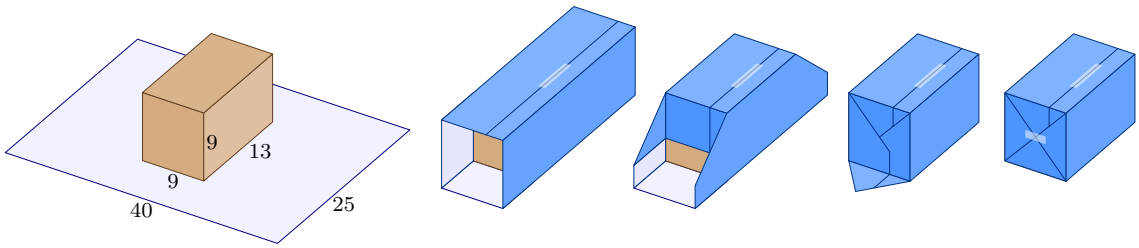


(A) 4 (B) 6 (C) 8 (D) 10 (E) 12

Questions 21 to 25, 5 marks each
21-25 题，每题 5 分

21. Amisha is wrapping a present for her little brother. The present is in a box measuring 9 cm × 9 cm × 13 cm and the sheet of wrapping paper is 40 cm × 25 cm. She wraps the present as tightly as possible, as shown.

小莎正在为她的弟弟包装礼物。礼物装在一个尺寸为 9 cm × 9 cm × 13 cm 的盒子里，然后用尺寸为 40 cm × 25 cm 的包装纸将盒子尽可能紧实地包裹好，如图所示。



What percentage of the wrapping paper is no longer visible from any angle?

请问从任何角度都看不见的那部分包装纸，占整张包装纸面积的百分之多少？

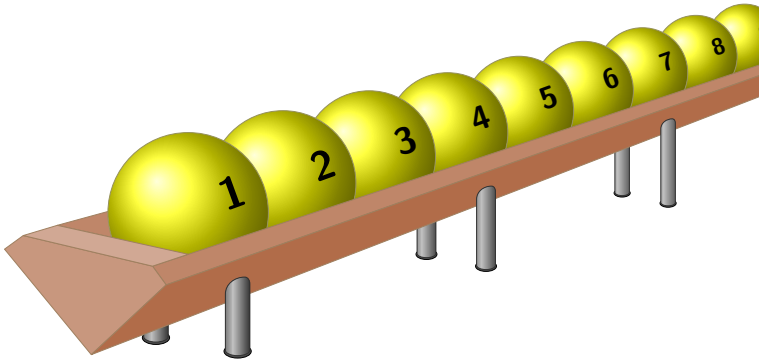
(A) 25% (B) 30% (C) 33% (D) 37% (E) 42%

---

22. A long line of ping-pong balls numbered 1, 2, 3, ... is sitting in order in a tilted chute, as shown below.
如图所示，一排编号为 1、 2、 3 ..... 的乒乓球按顺序排列在一个倾斜的滑槽中。

Xavier, Yasmin and Zebedee take turns removing balls from the chute. When a ball is removed, the rest roll down to fill the gap. When Zebedee has taken his ball, it is Xavier's turn again and they continue in the same order for as long as there are balls remaining.
小西、小雅和小泽轮流从滑槽中取球。当一个球被取出时，剩下的球会滚落填补空缺。当小泽取走他的球后，再次轮到小西取球。然后按照同样的顺序继续取球，直到没有剩余的球为止。

Xavier always takes the second available ball. 小西总是取滑槽中的第二个球。
Yasmin always takes the third available ball. 小雅总是取滑槽中的第三个球。
Zebedee always takes the first available ball. 小泽总是取滑槽中的第一个球。



Which statement is **true**?
请问下列哪项陈述是**正确**的？

(A) Xavier takes ball 48. 小西取走了编号为 48 的球。
(B) Yasmin takes ball 49. 小雅取走了编号为 49 的球。
(C) Zebedee takes ball 50. 小泽取走了编号为 50 的球。
(D) Yasmin takes ball 51. 小雅取走了编号为 51 的球。
(E) Xavier takes ball 52. 小西取走了编号为 52 的球。

23. To make the stamps easier to pull apart, sheets of postage stamps have rows of holes punched into them. In the sheet shown, each stamp has 7 holes along the top and bottom edges, 9 holes along the sides and 1 more at each corner. Holes are included around all four edges of the sheet so the borders can be torn off as well. The full sheet has 20 stamps arranged in 4 rows of 5. How many holes are there altogether?

版票上设有便于撕开邮票的孔。如图所示，每单张邮票的上下边各有 7 个孔，两侧各有 9 个孔，邮票的四个角处还各有 1 个孔。这些孔包括版票四周边缘的孔，以便边缘部分也能被撕下。整版邮票上共有 20 张单张邮票，每 5 张为一排，共 4 排。请问整版邮票上总共有多少个孔？

(A) 320 (B) 391 (C) 421 (D) 451 (E) 640

---

24. Sally arranges the numbers from 1 to 8 in a circle and then calculates the sum of each set of 3 adjacent numbers. She would like an arrangement in which the smallest of these sums is as large as possible. If she chooses such an arrangement, what will this smallest sum be?

小莉把数字 1 到 8 排成一个圈，然后以每 3 个相邻数字为一组，计算每组数字的总和。她期望找到一种排列方式，使得这些总和中的最小值尽可能大。如果按照小莉的想法排列，请问其中最小总和是多少？

(A) 10 (B) 11 (C) 12 (D) 13 (E) 14

---

25. A *totally even number* is a positive integer in which all the digits are even. The list of all totally even numbers begins as follows:

全偶数是指各位数字都是偶数的正整数。由所有全偶数组成的数列如下所示：

$$2, 4, 6, 8, 20, 22, 24, 26, 28, 40, \ldots$$

The number 40 is in the 10th position in this list. In which position is the totally even number 2024?

在这个数列中，40 排在第 10 位。请问全偶数 2024 排在该数列的第几位？

(A) 57 (B) 132 (C) 257 (D) 512 (E) 1012

For questions 26 to 30, shade the answer as an integer from 0 to 999 in the space provided on the answer sheet.

问题 26-30 的答案为 0-999 之间的整数，请将答案填涂在答题卡对应区域。

Questions 26–30 are worth 6, 7, 8, 9 and 10 marks, respectively.

第 26-30 题分别为 6, 7, 8, 9, 10 分。

26. Ringo has 5 cards with the numbers 2, 3, 4, 5 and 6 on them. He wrote a list of all possible two-digit numbers that can be made with these cards. Find the sum of all numbers in his list that are not prime.

小林有 5 张卡片，每张卡片上分别写有数字 2、3、4、5、6。他写下了用这些卡片可以组成的所有可能的两位数。请问小林列出的所有两位数中非质数的总和是多少？

---

27. A farmer, Jenni, grows oranges and grapefruit. At her last harvest she picked 120 grapefruit and a number of oranges. However, one-tenth of all grapefruit look like oranges and one-tenth of all oranges look like grapefruit. Overall, one-fifth of all fruits look like grapefruit. How many oranges did Jenni harvest?

农夫小詹种植了橙子和柚子。在最近的一次收获中，小詹收获了 120 个柚子和大量橙子。其中，有十分之一的柚子看起来像橙子，也有十分之一的橙子看起来像柚子。从整体上看，小詹收获的所有水果中有五分之一看起来像柚子。请问小詹收获了多少个橙子？

---

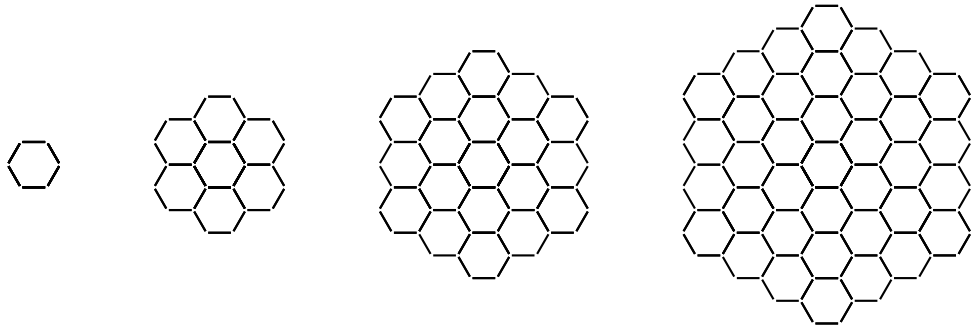
28. A rectangular piece of paper would lose $999\,\text{cm}^2$ of area if a $3\,\text{cm}$ wide border was trimmed from all four sides. If instead the trimmed border was $2\,\text{cm}$ wide, what area, in square centimetres, would be lost?

假设将一张长方形纸的四周剪去一个 $3\,\text{cm}$ 宽的边框，其面积会减少 $999\,\text{cm}^2$。如果将同一张纸的四周剪去一个 $2\,\text{cm}$ 宽的边框，请问其面积将会减少多少 cm^2？

---

29. Rodney builds increasingly large shapes out of unit rods. The smallest uses six rods and each larger shape adds a layer of hexagons around the previous shape.

小罗使用单位长度的棍子构建图形，且这些图形逐渐增大。最小的图形使用六根棍子，之后的每个图形都是通过在前一个图形周围加一层六边形构建而成的。



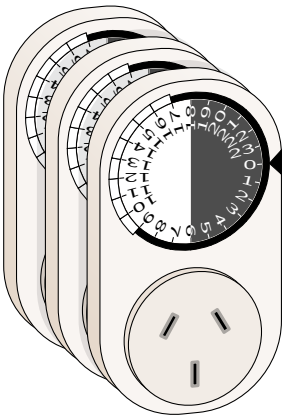
Starting with 2000 rods, Rodney builds a single shape like this, as large as possible. How many rods will he have left over?

小罗使用 2000 根棍子构建一个尽可能大的图形。请问他最后还能剩下多少根棍子？

---

30. Tim uses plug-in timers to automate his indoor plant lighting. Each timer has a plug on the back, a socket on the front, and a motorised dial that rotates once for every 24 hours that the plug has power.

小蒂用插电式定时器来实现室内植物补光灯自动化。每个定时器的背面都有插头，正面有插座，附带一个电动表盘。定时器插电后，电动表盘每 24 小时转一圈。



When the time on the dial is between 9:00 and 17:00, the timer switch is on, so that any power on the plug is switched through to the socket.

当表盘上的时间介于 9:00 到 17:00 之间时，定时器开启，此时插座接通电源。

For fun, he plugs together three of his timers, all set to time 0:00, with a lamp plugged into the front socket. How many hours until the lamp turns on?

小蒂做了一个有趣的实验。他将 3 个时间设置为 0:00 的定时器串联起来，并将一盏台灯插入到定时器正面的插座。请问需要多少个小时才能点亮这盏台灯？